Forward-Substitution-Based Generalized Eigenvalue Decomposition Processor for MU-MIMO Precoding

Chun-An Chen*, Chiao-En Chen†, and Yuan-Hao Huang*
*Institute of Communications Engineering and Department of Electrical Engineering, National Tsing Hua University, Taiwan
† Department of Electrical Engineering, National Chung Cheng University, Taiwan
E-mail:yhhuang@ee.nthu.edu.tw

***Abstract*—To improve the spectrum efficiency in wireless communication systems, multiple-input multiple-output (MIMO) technology uses multiple antennas and allows several users to share the same spectrum and antennas by using precoding technique. In the leakage-based precoding technique, generalized eigenvalue decomposition (GEVD) must generate many precoding matrices for all users in the base station to avoid co-channel interference. This paper presents a GEVD algorithm based on forward substitution (FS) scheme to avoid matrix inversion operations. This research also designed and implemented the GEVD processor by using a 40nm CMOS technology. The synthesis results show that the FS-based GEVD processor can reduce area cost by 52% and improve the processing throughput by 12% compared to our previous GEVD [1] processor based on triangular matrix inversion with block multiplication.**

I. Introduction

Multiple-input multiple-output (MIMO) technique can effectively increase the transmission capacity by using several antennas at both transmitter and receiver. In multi-user MIMO (MU-MIMO) scenario, a base station (BS) communicates with several user ends (UE) by time-division-multiple-access (TDMA) scheme or frequency-division-multiple-access (FDMA) scheme in traditional communication systems. In order to enhance the spectrum efficiency, the MU-MIMO system that combines the spatial-division-multiple-access (SDMA) and MIMO multiplexing has been included in the IEEE 802.11ac standard and is being considered in the 5G New Radio (NR) specifications. In the MU-MIMO system, a BS transmits multiple streams to multiple users spatially over the same bandwidth, but the transmitted signals suffer from co-channel interference (CCI) among users. Thus, transmitted signal vectors must be precoded according to the channel state information to suppress CCI. Tomlinson-Harashima precoder (THP) is a simple non-linear precoding scheme [2], and singular value decomposition (SVD) [3] and geometric mean decomposition (GMD) [4] [5] are popular linear precoding schemes. These precoding schemes are only designed for UEs with only one antenna.

The precoding schemes for UEs with multiple antennas can be categorized into block diagonalization (BD) [6] and leakage-based [7]–[9] precoding schemes. Because the BD-based scheme aims to perfectly cancel CCI at UEs without considering noise, it causes noise enhancement problem at low signal-to-noise ratio (SNR) condition. The leakage-based precoding scheme appropriately designs the precoder and decoder matrices by maximizing the signal-to-leakage-and-noise ratio (SLNR), thus leading to less noise enhancement. In the domain of generalized eigenspace, the leakage-based cost function satisfies SLNR constraint. Therefore, generalized eigenvalue decomposition (GEVD) can be used to generate the generalized eigenvectors of combined channels of all UEs in the leakage-based precoder/combiner.

Most of available precoding processors aimed to produce precoding matrices for UEs with single antenna. Our previous work [1] first designed and implemented a GEVD processor for UEs with multiple antennas based on a triangular matrix inversion algorithm with block-wise multiplication to avoid massive matrix inversion operations. This paper further proposes a new forward substitution (FS)-based GEVD algorithm to reduce the computational and hardware costs. This work also designed and implemented FS-based GEVD processor by using a 40nm CMOS technology.

The rest of this paper is organized as follows. Section II introduces the MU-MIMO system and the leakage-based precoding scheme based on the GEVD processing. Section III presents the proposed FS-based GEVD algorithm. Section IV shows the hardware architecture of the proposed GEVD processor. Section V demonstrates complexity analysis and IC synthesis results. Finally, Section VI concludes this paper.

II. GEVD for Multi-User MIMO Precoding

A. MU-MIMO System

Fig. 1 shows a MU-MIMO precoding communication system with N_T transmit antennas, N_R receive antennas, K users, and N_k receive antennas per user. The MIMO channel matrix $\mathbf{H} \in \mathbb{C}^{N_R \times N_T}$ consists of elements of independent and identically distributed (i.i.d) complex Gaussian random variables with $\mathcal{CN}(0,1)$. For the kth user, $N_k \times 1$ symbol vector $\tilde{\mathbf{x}}_k \in \mathbb{Q}^{N_k}$ is transmitted through an $N_k \times N_T$ channel $\mathbf{H}_k$ with complex Gaussian noise $\mathbf{n}_k \in \mathbb{C}^{N_k}$ with zero-mean and variance σ^2. In the transmitter, $\tilde{\mathbf{x}}_k$ is precoded by an $N_T \times N_k$ matrix $\mathbf{W}_k$. Finally, the transmitted symbols $\mathbf{x}$ is sum of $\mathbf{W}_k\tilde{\mathbf{x}}_k$. Therefore, the kth user received the signal

$$\mathbf{y}_k = \mathbf{H}_k\mathbf{W}_k\tilde{\mathbf{x}}_k + \mathbf{H}_k \sum_{l=1,l\neq k}^{K} \mathbf{W}_l\tilde{\mathbf{x}}_l + \mathbf{n}_k, \text{ where } k = 1, ..., K, \quad (1)$$

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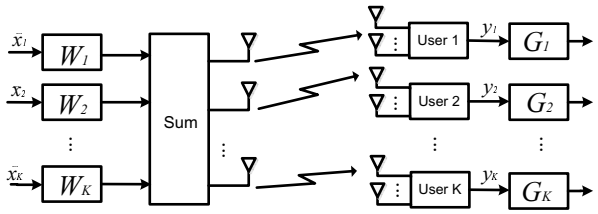


Fig. 1. The MU-MIMO system model.

where the second term is the CCI caused by other UEs. The precoders $\mathbf{W}_k$ and decoders $\mathbf{G}_k \in \mathbb{C}^{N_k \times N_k}$ must be designed to reduce the CCI from other UEs in the MU-MIMO system.

B. Leakage-Based Precoder

The leakage-based precoding [7] [8] [9] [10] maximizes signal-to-leakage-noise ratio (SLNR) for all users to determine $\mathbf{W}_k$ and $\mathbf{G}_k$. The SLNR of the kth user is expressed as

$$\begin{aligned} SLNR_k &= \frac{\|\mathbf{H}_k\mathbf{W}_k\|^2}{N_k\sigma^2 + \sum_{l=1,l\neq k}^{K}\|\mathbf{H}_l\mathbf{W}_k\|^2} \\ &= \frac{tr\{\mathbf{W}_k^H\mathbf{H}_k^H\mathbf{H}_k\mathbf{W}_k\}}{tr\{\mathbf{W}_k^H(\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k + \sigma^2\mathbf{I}_{N_T})\mathbf{W}_k\}}, \end{aligned} \tag{2}$$

where

$$\bar{\mathbf{H}}_k = [\mathbf{H}_1^T, ..., \mathbf{H}_{k-1}^T, \mathbf{H}_{k+1}^T, ..., \mathbf{H}_K^T]^T \ \in \mathbb{C}^{(N_r - N_k)\times N_T}, \tag{3}$$

and $\mathbf{H}_k^H\mathbf{H}_k \in \mathbb{C}^{N_T\times N_T}$ is a Hermitian matrix. $(\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k + \sigma^2\mathbf{I}_{N_T}) \in \mathbb{C}^{N_T\times N_T}$ is a Hermitian and positive definite matrix. Defined by generalized eigenspaces, precoding matrix $\tilde{\mathbf{W}}_k$ satisfies

$$\begin{aligned} \tilde{\mathbf{W}}_k^H\mathbf{H}_k^H\mathbf{H}_k\tilde{\mathbf{W}}_k &= \mathbf{D}_k \\ \tilde{\mathbf{W}}_k^H(\bar{\mathbf{H}}_k^H\mathbf{H}_k + \sigma^2\mathbf{I}_{N_T})\tilde{\mathbf{W}}_k &= \mathbf{I}, \end{aligned} \tag{4}$$

where $\mathbf{D}_k$ is a diagonal matrix with generalized eigenvalues of $(\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k + \sigma^2\mathbf{I}_{N_T})$ and $\mathbf{H}_k^H\mathbf{H}_k$ in descending order of their values. Thus, $\tilde{\mathbf{W}}_k$ is the generalized eigenmatrix. Note that the generalized eigenvectors of $(\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k+\sigma^2\mathbf{I}_{N_T})$ and $\mathbf{H}_k^H\mathbf{H}_k$ are equivalent to eigenvectors of $(\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k+\sigma^2\mathbf{I}_{N_T})^{-1}\mathbf{H}_k^H\mathbf{H}_k$. Substituting Equation (4) into (2), the denominator of SLNR is related with sum of the generalized eigenvalues. Generalized eigenvectors corresponding to larger eigenvalues must be selected in the precoder $\mathbf{W}_k$. To meet the criterion of leakage-based precoding scheme, the precoder can be given by

$$\mathbf{W}_k = \rho\tilde{\mathbf{W}}_k \begin{bmatrix} \mathbf{I}_{N_k} \\ \mathbf{0} \end{bmatrix}, \tag{5}$$

where $\tilde{\mathbf{W}}_k \in \mathbb{C}^{N_T\times N_T}$ is generalized eigenmatrix of $\mathbf{H}_k^H\mathbf{H}_k$ and $(\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k+\sigma^2\mathbf{I}_{N_T})$. The concatenation of $N_k\times N_k$ identity matrix $\mathbf{I}_{N_k}$ and zero matrix $\mathbf{0}$ selects the leading N_k columns of eigenmatrix $\tilde{\mathbf{W}}_k$ corresponding to N_k larger eigenvalues. ρ is the power constraint to ensure

$$tr\{\mathbf{W}_k^H\mathbf{W}_k\} = N_k. \tag{6}$$

Finally, the decoder of the kth user can be obtained by

$$\mathbf{G}_k = \frac{\mathbf{W}_k^H\mathbf{H}_k^H}{\|\mathbf{H}_k\mathbf{W}_k\|_F}. \tag{7}$$

With $\mathbf{W}_k$ precoding and $\mathbf{G}_k$ decoding, the received signal at the kth user can be approximated by

$$\mathbf{G}_k\mathbf{y}_k \approx \alpha\mathbf{D}_k\tilde{\mathbf{x}}_k + \mathbf{v}_k. \tag{8}$$

Notice that the second ICC term of Equation (1) could be approximately eliminated by suppressing leakage and noise.

C. Generalized Eigenvalue Decomposition

To realize the GEVD process, $\mathbf{Q}_k$ and $\mathbf{P}_k$ are, respectively, defined by

$$\mathbf{Q}_k = \mathbf{H}_k^H\mathbf{H}_k \ , \ \mathbf{P}_k = \bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k + \sigma^2\mathbf{I}_{N_T}, \tag{9}$$

where both $\mathbf{Q}_k \in \mathbb{C}^{N_T\times N_T}$ and $\mathbf{P}_k \in \mathbb{C}^{N_T\times N_T}$ are Hermitian and $\mathbf{P}_k$ is positive definite. According to Equation (4), $\mathbf{Q}_k$ is transformed to a diagonal matrix, and $\mathbf{P}_k$ is transformed to an identity matrix by the optimal precoding matrix $\tilde{\mathbf{W}}_k$, which can be generated by the GEVD algorithm with four basic steps [11]. The first step is the Cholesky decomposition of $\mathbf{P}_k$

$$\mathbf{P}_k = \mathbf{G}_k\mathbf{G}_k^H. \tag{10}$$

The second step generates a matrix $\mathbf{A}_k$ using $\mathbf{G}_k$ and $\mathbf{Q}_k$ by

$$\mathbf{A}_k = \mathbf{G}_k^{-1}\mathbf{Q}_k(\mathbf{G}_k^{-1})^H. \tag{11}$$

Afterwards, the third step finds a unitary matrix $\mathbf{V}_k$ to diagonalize $\mathbf{A}_k$. Several methods, such as symmetric QR and Jocobi algorithms, can be used to produce $\mathbf{V}_k$ from $\mathbf{A}_k$ Finally, the generalized eigen matrix $\tilde{\mathbf{W}}_k$ is calculated by

$$\tilde{\mathbf{W}}_k = (\mathbf{G}_k^{-1})^H\mathbf{V}_k. \tag{12}$$

The generic GEVD algorithm consists of a great amount of matrix inversion calculations. Therefore, this paper presents a low-complexity GEVD algorithm using forward substitution to prevent matrix inversion calculations.

III. Forward Substitution-Based GEVD Algorithm

A. Cholesky Decomposition

Cholesky decomposition is a special case of LU decomposition. If $\mathbf{P}$ that is a Hermitian and positive-definite matrix, $\mathbf{P}$ can be factorized into the product of a matrix and its conjugate transpose matrix by Cholesky decomposition as follows:

$$\mathbf{P} = \mathbf{G}\mathbf{G}^H, \tag{13}$$

where $\mathbf{G}$ is a lower triangular matrix and its diagonal elements are positive. In order to compute $\mathbf{G}$ from $\mathbf{P}$, we can expand Equation (13) as follows:

$$\begin{aligned} \mathbf{P}(i,j) &= \mathbf{G}(i,1)\mathbf{G}^H(1,j) + \cdots + \mathbf{G}(i,N_T)\mathbf{G}^H(N_T,j) \\ &= \mathbf{G}(i,1)\mathbf{G}^H(1,j) + \cdots + \mathbf{G}(i,j)\mathbf{G}^H(j,j) \\ &= \mathbf{G}(i,1)(\mathbf{G}(j,1))^* + \cdots + \mathbf{G}(i,j)(\mathbf{G}(j,j))^*, \end{aligned} \tag{14}$$

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where $i, j \in \{1, 2, \cdots, N_T\}$. Since $\mathbf{G}^H$ is a upper triangular matrix, the elements of $\mathbf{G}^H$ are zero when $i > j$. Equation (14) can be arranged by

$$\mathbf{P}(:, j) = \sum_{k=1}^{j} (\mathbf{G}(j,k))^* \mathbf{G}(:,k). \tag{15}$$

By rearranging Equation (15), $\mathbf{G}$ can be expressed by $\mathbf{P}$ column by column as follows:

$$(\mathbf{G}(j,j))^* \mathbf{G}(:,j) = \mathbf{P}(:,j) - \sum_{k=1}^{j-1} (\mathbf{G}(j,k))^* \mathbf{G}(:,k) \equiv \mathbf{v}. \tag{16}$$

Algorithm 1 shows the Cholesky decomposition algorithm.

Algorithm 1: Cholesky Decomposition

INPUT : $\mathbf{P}$
OUTPUT : $\mathbf{G}$
1 : **for** $j = 1 : \mathrm{N_T}$
2 : $\quad \mathbf{v}(j : N_T) = \mathbf{P}(j : N_T, j)$
3 : $\quad$ **for** $k = 1 : \mathrm{j} - 1$
4 : $\quad \mathbf{v}(j : N_T) = \mathbf{v}(j : N_T) - \mathbf{G}(j,k)\mathbf{G}(j : N_T, k)$
5 : $\quad$ **end**
6 : $\quad \mathbf{G}(j,j) = \sqrt{\mathbf{v}(j)}$
7 : $\quad \mathbf{G}(j : N_T, j) = \frac{\mathbf{v}(j:N_T)}{\mathbf{G}(j,j)}$
8 : **end**

B. Jacobi Method

The proposed algorithm uses Jacobi method [11] to diagonalize matrix $\mathbf{A}_k$. For complex $\mathbf{Q}_k$, the elements (i,i), (j,j), (i,j), and (j,i) of the rotation matrix $\mathbf{J}$ are expressed by $\cos\theta e^{j\phi}$, $\cos\theta$, $\sin\theta e^{j\phi}$, and $-\sin\theta$, respectively.

$$\mathbf{J}(i,j,\theta,\phi) = \begin{bmatrix} 1 & & & & & & 0 \\ & \ddots & & & & & \\ & & \cos\theta e^{j\phi} & & \sin\theta e^{j\phi} & & \\ & & & \ddots & & & \\ & & -\sin\theta & & \cos\theta & & \\ & & & & & \ddots & \\ 0 & & & & & & 1 \end{bmatrix}, \tag{17}$$

where θ and ϕ are, respectively, obtained by $\tan(2\theta) = \frac{2|\mathbf{A}_{ij}|}{\mathbf{A}_{jj} - \mathbf{A}_{ii}}$ and $e^{j\phi} = \frac{\mathbf{A}_{ij}}{|\mathbf{A}_{ij}|}$. The elements (i,j) and (j,i) of $\mathbf{A}$ become zero in one rotation. Jacobi method is an iterative algorithm the converts $\mathbf{A}$ to a diagonal matrix through many times rotation operations. A constant *sweep* denotes the rotation times for zeroing all off-diagonal elements of $\mathbf{A}$.

C. Forward Substitution

The proposed algorithm utilizes forward substitution to calculate the matrix $\mathbf{A}$ of Equation (11), which is first modified by multiplying $\mathbf{G}_k$ on both sides.

$$\mathbf{G}_k \mathbf{A}_k = \mathbf{Q}_k (\mathbf{G}_k^{-1})^H = \varphi_k, \tag{18}$$

where $\varphi_k \in \mathbb{C}^{N_T \times N_T}$. Secondly, this equation can be further multiplied by $\mathbf{G}_k^H$ on both sides as follows:

$$\mathbf{Q}_k = \varphi_k \mathbf{G}_k^H. \tag{19}$$

These two matrix multiplications are so called the forward substitution, which is used to avoid the matrix inversion in Equation (11). Because $\mathbf{G}_k^H$ is an upper triangular matrix, the matrix φ_k can be solved after $\mathbf{Q}_k$ is given. If $N_T = 8$, Equation (19) is expressed by

$$\mathbf{Q} = \begin{bmatrix} \phi_{11} & \phi_{12} & \cdots & \phi_{1N_T} \\ \phi_{21} & \phi_{22} & & \phi_{2N_T} \\ \vdots & & \ddots & \vdots \\ \phi_{N_T 1} & \phi_{N_T 2} & \cdots & \phi_{N_T N_T} \end{bmatrix} \begin{bmatrix} g_{11}^{\mathrm{H}} & g_{12}^{\mathrm{H}} & \cdots & g_{1N_T}^{\mathrm{H}} \\ 0 & g_{22}^{\mathrm{H}} & & \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 0 & g_{N_T N_T}^{\mathrm{H}} \end{bmatrix} \tag{20}$$

Thus, the matrix φ_k can be obtained by iteratively solving the jth column of matrix φ_k from the previous $j-1$ solved columns as follows:

$$(\mathbf{G}(j,j))^H \varphi(:,j) = \mathbf{Q}(:,j) - \sum_{k=1}^{j-1} (\mathbf{G}(k,j))^H \varphi(:,k), \tag{21}$$

where $(\mathbf{G}(j,j))^H = \mathbf{G}(j,j)$. Because the diagonal components of $\mathbf{G}$ are positive. Thus, the jth column of φ_k can be expressed by

$$\varphi(:,j) = (\mathbf{Q}(:,j) - \sum_{k=1}^{j-1} (\mathbf{G}^H(k,j))\varphi(:,k))/\mathbf{G}(j,j), \tag{22}$$

After φ_k is obtained, Equation (18) is modified as follows:

$$\varphi_k^H = \mathbf{A}_k^H \mathbf{G}_k^H = \mathbf{A}_k \mathbf{G}_k^H. \tag{23}$$

Thus, the matrix $\mathbf{A}_k$ can be also solved as Equation (19) and Equation (20). Because $\mathbf{G}_k^H$ is upper triangular, Equation (23) can be solved by using the same iterative operations as solving φ_k. Thus, two forward substitutions can share the same hardware circuit and support course-grained pipeline architecture in the hardware implementation for producing $\mathbf{A}_k$ and φ_k. Finally, Equation (12) also needs to be modified as follows because $(\mathbf{G}_k^{-1})^H$ is not produced in the forward substitution.

$$\mathbf{V}_k = \mathbf{G}_k^H \tilde{\mathbf{W}}_k. \tag{24}$$

Since the matrix $\tilde{\mathbf{W}}_k$ is solved from the last to the first column, this calculation is called backward substitution, which is used to generate precoding matrix $\tilde{\mathbf{W}}_k$. The complete forward substitution method can be expressed as Algorithm 2.

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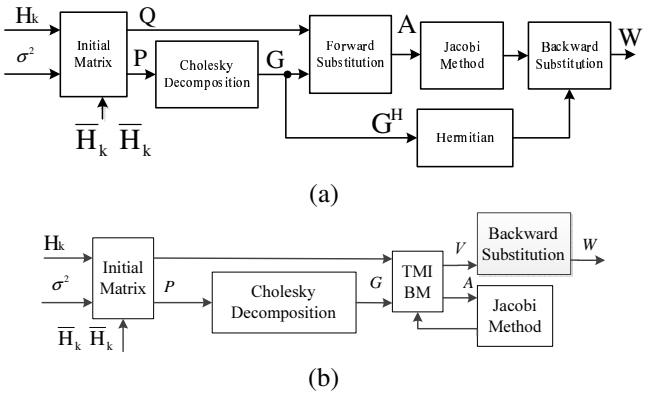


Fig. 2. The GEVD processor architectures based on (a) forward substitution and (b) triangular matrix inversion with block-wise multiplication [1].

Algorithm 2: Forward Substitution

INPUT : $\mathbf{G}, \mathbf{Q}$
OUTPUT : $\mathbf{A}$

for $j = 1 : \mathrm{N_T}$
  $\mathbf{v} = \mathbf{Q}(:, j)$
  for $k = 1 : \mathrm{j} - 1$
    $\mathbf{v} = \mathbf{v} - \varphi(:, k)\mathbf{G}^{\mathrm{H}}(k, j)$
  end
  $\varphi(:, j) = \frac{\mathbf{v}}{\mathbf{G}^{\mathrm{H}}(j,j)}$
end
for $j = 1 : \mathrm{N_T}$
  $\mathbf{v} = \varphi^{\mathrm{H}}(:, j)$
  for $k = 1 : \mathrm{j} - 1$
    $\mathbf{v} = \mathbf{v} - \mathbf{A}(:, k)\mathbf{G}^{\mathrm{H}}(k, j)$
  end
  $\mathbf{A}(:, j) = \frac{\mathbf{v}}{\mathbf{G}^{\mathrm{H}}(j,j)}$
end

IV. Processor Architecture

A. System Architecture

Fig. 2 shows the GEVD architecture based on forward substitution. This processor generates a precoder matrix W_k for the k-th user. The forward substitution and backward substitution are used to realize Step 2 and 4, respectively. Figure 2 shows that the blocks, initial matrix, Cholesky decomposition, and Jacobi method, of the FS-based GEVD processor are the same as those in the TMI-BM-based GEVD processor of our previous work [1].

1) Matrix Initial: The matrix initialization block calculates $\mathbf{H}_k^H\mathbf{H}_k$ and $\bar{\mathbf{H}}_k^H\bar{\mathbf{H}}_k + \sigma^2\mathbf{I}_{N_T}$ as the GEVD inputs with a 8×4 by 4×1 matrix multiplier, which are realized by eight 4 by 4 vector multiplications because the following Cholesky decomposition operates column by column.

2) Cholesky Decomposition: Figure 3 shows the architecture of Cholesky decomposition, which iteratively calculates the columns of matrix $\mathbf{G}$. The register file $\mathbf{G}$ is stored as the inputs of scalar-vector multipliers. The adder network computes the second term of Equation (16).

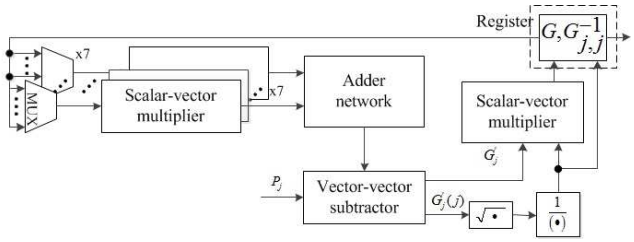


Fig. 3. The architecture of Cholesky decomposition.

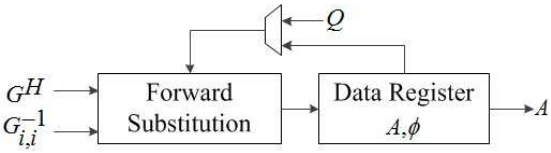


Fig. 4. The architecture of forward substitution processing.

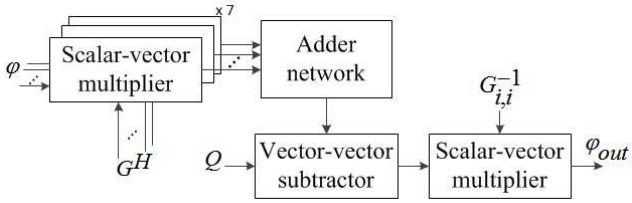


Fig. 5. Forward substitution architecture.

3) Forward Substitution: Fig. 5 shows the architecture of forward substitution, which generates matrix φ and $\mathbf{A}$ by computing the first column to the last column of them. Firstly, matrix $\mathbf{Q}$ is used as input to generate the matrix φ. Afterwards, the resultant φ of the first forward substitution is used as input to generate the matrix $\mathbf{A}$. Other inputs such as $\mathbf{G}^H$ and as $\mathbf{G}_{i,i}^{-1}$ are from Cholesky decomposition in Fig. 2. The forward substitution circuit consists of seven parallel scalar-vector multipliers, one adder network, one vector-vector subtractor, and one scalar-vector multiplier. For $N_T = 8$, at most seven parallel scalar-vector multipliers are required to compute the eighth column from the previous seven columns. The output of adder network corresponds to second term in Equation (21). The architecture of backward substitution is the same as that of forward substitution except that the input is the Jacobi method-rotated matrix and column input sequence is reversely ordered.

4) Jacobi Method: Figure 6 shows the Jacobi processing architecture. $sweep = 2$ is used to fix the rotation processing time, and Jacobi algorithm is unfolded to two rotation blocks. Each Jacobi rotation block has four stages as shown in Figure 7. Stage 1 and Stage 2 rotate columns of $\mathbf{A}_k$ while Stage 3 and Stage 4 rotate rows of $\mathbf{A}_k$. Jacobi rotation matrix contains additional $e^{j\phi}$ component. Thus, Stage 1 and Stage 3 rotates the ith column and ith row with ϕ and $-\phi$, respectively. Then, Stage 2 and Stage 4 rotates columns and rows with angle θ. Because Jacobi rotation block is unfolded by two, Stage 3

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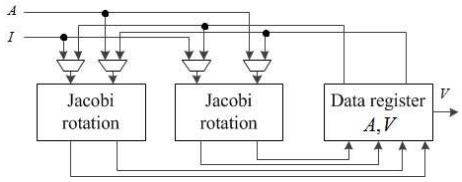


Fig. 6. The architecture of Jacobi method.

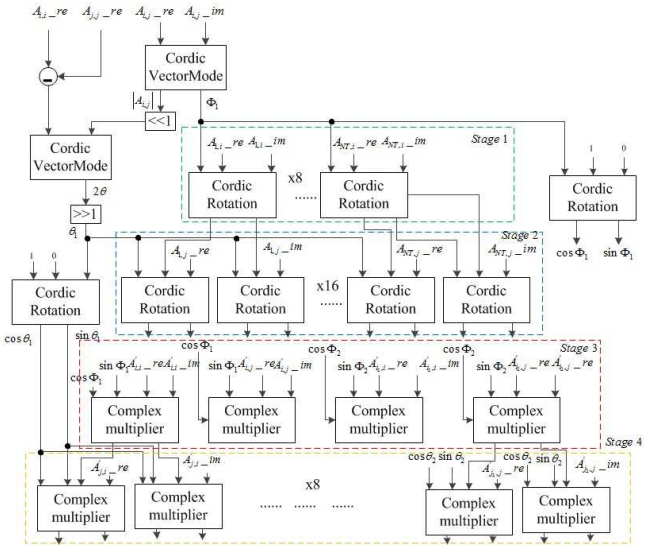


Fig. 7. The architecture of Jacobi rotation for A.

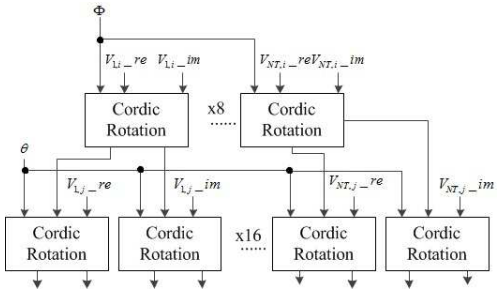


Fig. 8. The architecture of Jacobi rotation for V.

and Stage 4 needs to rotate i_2th and j_2th rows with ϕ_2 and θ_2, respectively, by the other Jacobi rotation block. Jacobi rotation is implemented by CORDIC circuits and complex multipliers. CORDICs are utilized to rotate matrix and calculate $\cos\theta$ and $\sin\theta$ in Jacobi rotation in Step 1 and 2. Stage 3 and Stage 4 are implemented by complex multipliers. Figure 8 shows the architecture of Jacobi rotation for $\mathbf{V}$. The initial matrix for calculating $\mathbf{V}$ is an identity matrix $\mathbf{I}$. The CORDIC circuits are the same as Stage 1 and Stage 2 circuits for calculating $\mathbf{A}$ matrix, and θ and Φ in Figure 8 are obtained from CORDIC rotation for $\mathbf{A}$ in Figure 7.

V. Experiments

A. Complexity Analysis

Table I compares the computational complexity of the TMI-BM-based and FS-based GEVD algorithms. The complexity

TABLE I
COMPUTATIONAL AND ARCHITECTURAL COSTS OF THE TMI-BM AND FS SCHEMES.

Computational Cost		
	TMI-BM	FS
TMI	$\frac{1}{3}\mathrm{N_T^3}$	0
BM	$\frac{2}{3}\mathrm{N_T^3}$	0
φ	0	$\frac{1}{2}\mathrm{N_T^3}$
$\mathbf{A}$	0	$\frac{1}{6}\mathrm{N_T^3}$
Total	$\mathrm{N_T^3}$	$\frac{2}{3}\mathrm{N_T^3}$
Architectural Cost		
	TMI-BM	FS
Clock cycle	$\log_2 N_T + 2$	$2N_T$
# of multipliers	$\frac{3}{4}N_T^3$	N_T^2

TABLE II
TOTAL COMPUTATIONAL COMPLEXITY OF THE GEVD PROCESSING WITH THE TMI-BM AND FS SCHEMES. (T: SWEEP TIMES)

	TMI-BM	FS
Step 1	$\frac{N_T^2+N_T}{2}N_k + \frac{2}{3}N_T^3$	$\frac{N_T^2+N_T}{2}N_k + \frac{2}{3}N_T^3$
Step 2	N_T^3	$\frac{2}{3}N_T^3$
Step 3	$8t(N_T+1)\frac{N_T^2-N_T}{2}$	$8t(N_T+1)\frac{N_T^2-N_T}{2}$
Step 4	$\frac{1}{2}N_T^3$	$\frac{1}{2}N_T^3$

of the FS scheme is $\frac{1}{3}{\mathrm{N_T}}^3$ less than that of the TMI-BM scheme because the matrix inversion of $\mathbf{G}$ is not required in the FS scheme. Although the TMI-BM scheme can avoid the matrix inversion by block multiplication, the dimension of the triangular matrix inversion increases along with the iteration, leading to the slight increase of complexity along with the transmission antenna number. Table II shows the overall complexity of the GEVD processing of the TMI-BM and FS schemes in terms of complex multiplication. Step 1 includes the processing of initial matrix and Cholesky decomposition. Step 2 consists of the TMI-BM scheme or forward substitution scheme. Step 3 and 4 are the Jacobi method and backward substitution processing, respectively. Considering more about the hardware issue, Table I also estimates the time and hardware costs by showing the cycle counts and multiplier numbers of two schemes when we use their intrinsic degrees of parallelism [12]. The TMI-BM scheme shows a lower latency than the FS scheme, but the FS scheme achieves lower hardware cost than the TMI-BM scheme when the number of transmitted antennas increases in the MIMO system. There, the FS scheme is more suitable for the precoding processing in the massive MIMO system.

B. VLSI Implementation

The proposed GEVD processor was adopted the BLR-SLNR-THP precoding technique [10] to verify the BER performance in a two-user 8×8 MU-MIMO system with $N_k = 4$ and $K = 2$. Fig. 9 compares the BER performances of the fixed-point and floating-point GEVD processing. The FS-based GEVD processor was implemented by using TSMC 40nm CMOS technology and the synthesis results were compared with the TMI-BM-based GEVD processor [1] and

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TABLE III
COMPARISON OF MIMO PRECODING PROCESSOR CHIPS.

	This work	[1]	[2]	[3]	[4]	[5]	
Algorithm	FS GEVD	TMI-BM GEVD	LR-THP	SL-SVD	GMD	EVD	GMD
Technology	40nm	40nm	90nm	90nm	90nm	40nm	40nm
MIMO $N_T \times N_k$	8×4	8×4	4×1	4×1	4×1	4×1	4×1
# of users K	2	2	4	4	4	4	4
Max Freq. (Hz)	87MHz	73M	120M	182M	170M	120	120
Area	$3.7mm^2$	$7.7mm^2$	$0.5mm^2$				
Gate Count				120k	415k		
Throughput (Mat/sec)	0.515M	0.46M	2.6M	7M	42.5M	0.99M	2.96M
Nor. Throughput (Mat/sec)	117.2M	104.7M	2.6M	7M	42.5M	0.99M	2.96M

Nor. Throughput = Throughput $\times \frac{(\sqrt{N_T \times N_k \times K})^3 \times {N_k}^3 \times Technology}{\sqrt{4 \times 1 \times 4}^3 \times 1^3 \times 90nm}$

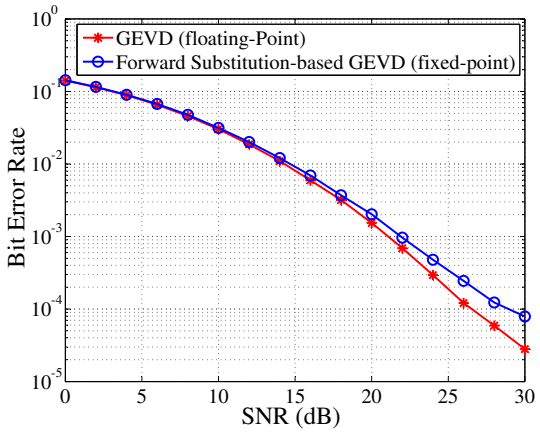


Fig. 9. Fixed point simulation results.

other works, as shown in Table III. Notice that the proposed precoding processor must not only cancel the interference among UEs but also eliminate the interference among antennas for each UE. The previous work only supports one antenna in the UE and needs only to cancel the user interference. In the comparison table, the chip throughputs were normalized by matrix size, IC technology, and the receive antenna number N_k. When compared with the TMI-BM-based GEVD processor, the FS-based GEVD processor greatly reduces the hardware cost by 52% because the FS scheme uses parallel scalar-vector and vector-vector multipliers while the TMI-BM scheme uses parallel matrix-matrix multipliers. Although the latency of the FS scheme is larger than that of the TMI-BM scheme, they occupy only a small portion of the overall latency of the GVED processing. Thus, the overall throughput of the FS-based GEVD processor, with a faster clock speed, is a higher than that of the TMI-BM-based GEVD processor by 12%.

VI. CONCLUSION

This paper presents a GEVD algorithm for the leakage-based MU-MIMO precoding technique. The proposed algorithm avoids the massive matrix inversion processing by successive forward substitution of matrices and uses smaller gains of scaler-vector/vector-vector multiplications to reduce the computation and hardware cost. Synthesis results show that the proposed FS-based GEVD processor can reduce more than half chip area with slightly throughput improvement.

REFERENCES

[1] C.-A. Chen, Z.-F. Yang, C.-E. Chen, and Y.-H. Huang, "A generalized eigenvalue decomposition processor for multi-user mimo precoding," in *IEEE Asian Pacific Conference on Circuits and Systems*, 2016, pp. 281–284.

[2] C.-F. Liao, F.-C. Lan, J.-W. Jhang, and Y.-H. Huang, "A 576-mbit/s 64-qam 4 4 mimo precoding processor with lattice reduction," *Circuits and Systems II: Express Briefs, IEEE Transacrion on*, vol. 61, no. 2, pp. 95–99, Feb. 2014.

[3] C.-Z. Zhan, Y.-L. Chen, and A.-Y. Wu, "Iterative Superlinear-Convergence SVD Beamforming Algorithm and VLSI Architecture for MIMO-OFDM Systems," *Signal Processing, IEEE Transacrion on*, vol. 60, no. 6, pp. 3264–3277, June 2012.

[4] Y.-T. Hwang, W.-D. Chen, and C.-R. Hong, "A Low Complexity Geometric Mean Decomposition Computing Scheme and Its High Throughput VLSI Implementation," *Circuits and Systems I: Regular Papers, IEEE Transacrion on*, vol. 61, no. 3, pp. 1170–1182, April 2014.

[5] Y. Wu and P. Y. Tsai, "A generalized matrix-decomposition processor for joint mimo transceiver design," in *in Proc. of 2017 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2017, pp. 1153–1157.

[6] Keke Zu; de Lamare, R.C.; Haardt, M., "Generalized Design of Low-Complexity Block Diagonalization Type Precoding Algorithms for Multiuser MIMO Systems," vol. 57, no. 4, pp. 4232–4242, October 2013.

[7] S. Zhao, Y. Zhao, H. Sun, J. Liu, L. Zhang, and L. Gui, "An interference-aware precoding scheme with other-cell interference for downlink milti-user mimo channel," in *IEEE Vehicular Technology Conference*, vol. 57, no. 4, Sept 2010.

[8] U. Jayasinghe, N. Rajatheva, and M. Latva-aho, "Leakage based multi user beamforming scheme to mitigate interference in mimo-fbmc," in *17th International ITG Workshop on Smart Antennas*, vol. 57, no. 4, March 2013.

[9] Sadek, M.; Tarighat, A.; Sayed, A.H., "A Leakage-Based Precoding Scheme for Downlink Multi-User MIMO Channels," vol. 57, no. 4, pp. 1711–1721, May 2007.

[10] C.-E. Chen, T.-W. Cho, and W.-H. Chung, "Blockwise-lattice-reduction-aided Tomlinsion-Harashima precoder Designs for MU-MIMO downlink communications with clusters of correlated users," vol. 57, no. 4, Mar 2014.

[11] H.Stark and J.W.Woods, "Matrix computations," in *Edition Prentice Hall*, 2002.

[12] G. G. Lee, H.-Y. Lin, C.-F. Chen, and T.-Y. Huang, "Quantifying intrinsic parallelism using linear algebra for algorithm/architecture co-exploration," *IEEE Transactions on Parallel and Distributed Systems*, vol. 23, no. 5, pp. 944–957, May 2012.

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